

CONROE-NORTH HOUSTON REGIONAL, TX, USA

WMO: 722444

Lat: 30.357N

Lon: 95.414W

Elev: 245

StdP: 14.57

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 53902

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.5	29.9	15.9	12.4	41.3	20.2	15.3	43.3	21.1	58.8	19.4	60.6	2.2	340	0.381

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.3	97.8	75.3	95.7	75.9	93.5	76.0	79.8	88.1	79.0	87.6	78.3	87.1	7.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.0	146.6	82.3	77.0	142.0	81.9	76.2	137.9	81.5	43.3	87.6	42.6	87.5	41.9	87.7	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.8	16.9	15.2	DB	21.8	100.9	3.8	3.0	19.0	103.1	16.8	104.8	14.6	106.5	11.8	108.7	(4)
(5)			WB	20.4	81.3	3.4	1.3	18.0	82.2	16.0	83.0	14.1	83.7	11.6	84.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	68.0	50.6	54.8	61.3	68.0	74.9	80.7	82.6	83.3	78.4	69.1	59.4	52.2		
	DBStd	13.69	9.66	10.09	8.80	7.08	5.64	3.18	2.85	3.20	4.83	7.71	9.23	9.45		
	HDD50	291	110	59	14	1	0	0	0	0	0	1	22	84		
	HDD65	1679	457	310	178	51	5	0	0	0	1	48	216	413		
	CDD50	6863	129	193	364	540	772	922	1010	1032	851	593	304	153		
	CDD65	2777	12	24	62	140	312	472	545	567	402	176	48	17		
	CDH74	27787	39	108	383	1092	2881	4924	6078	6443	3850	1620	310	59		
	CDH80	12590	0	9	49	247	1076	2361	3034	3422	1802	541	47	2		
(14) Wind	WSAvg	5.6	6.2	6.7	6.9	7.2	6.6	5.2	4.6	4.4	4.0	4.6	5.3	5.8		
(15) Precipitation	PrecAvg	47.0	4.0	3.1	3.3	3.1	4.1	5.2	3.3	2.8	3.6	5.1	4.5	4.0		
	PrecMax	66.8	10.3	6.2	8.1	10.3	11.1	16.5	10.7	7.9	7.2	21.3	14.3	8.8		
	PrecMin	25.7	0.9	0.5	0.2	0.2	0.0	0.6	0.4	0.0	1.0	0.1	0.5	1.1		
	PrecStd	9.2	2.3	1.9	1.8	2.3	3.1	3.4	2.3	2.2	1.8	5.0	3.0	1.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.3	81.3	84.1	88.6	93.3	98.5	99.2	101.3	99.7	91.1	84.2	79.2		
		MCWB	65.5	66.9	68.1	71.6	74.4	74.8	75.5	74.6	72.9	74.3	72.5	69.2		
	2%	DB	74.5	77.4	81.2	85.2	90.6	95.2	96.6	98.4	94.6	88.8	81.1	75.2		
		MCWB	64.3	66.1	67.7	70.0	74.1	75.4	75.8	75.6	74.3	73.7	69.4	67.1		
	5%	DB	71.6	73.9	78.7	82.5	88.5	92.9	94.4	96.2	91.7	85.9	77.7	72.4		
		MCWB	63.8	64.9	66.4	69.4	74.0	75.9	76.3	76.3	74.0	71.9	67.1	66.4		
	10%	DB	68.2	70.9	75.2	80.7	86.3	90.8	92.5	93.7	89.8	82.6	74.4	69.7		
		MCWB	61.9	64.2	64.6	68.4	73.3	75.9	76.6	76.4	74.1	70.0	66.9	64.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.7	70.9	72.4	75.7	78.7	80.4	80.5	80.5	79.6	78.5	75.1	71.7		
		MCDB	73.5	74.9	79.3	82.6	87.8	88.0	88.0	89.2	86.7	84.6	79.8	75.2		
	2%	WB	67.5	69.2	70.8	74.0	77.4	79.2	79.5	79.6	78.4	77.1	72.5	70.1		
		MCDB	71.3	74.2	76.4	80.7	85.9	87.4	87.9	88.7	86.0	83.9	77.2	73.6		
	5%	WB	66.0	67.6	69.4	72.5	76.2	78.2	78.7	78.9	77.7	75.5	70.9	68.3		
		MCDB	70.3	72.0	74.3	78.9	84.1	86.8	87.3	88.2	85.5	81.5	75.3	71.8		
	10%	WB	63.2	65.7	68.1	71.3	75.2	77.4	78.1	78.2	76.7	73.9	69.0	65.6		
		MCDB	67.0	69.8	72.8	77.2	82.5	86.2	86.9	87.6	84.4	79.4	73.3	68.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	22.0	21.3	22.3	22.0	20.6	19.8	19.6	21.3	21.5	23.8	22.9	21.6		
		MCDBR	24.1	22.8	22.4	22.7	22.0	23.0	22.6	24.4	24.6	24.3	23.6	23.5		
		MCWBR	15.2	13.3	10.9	10.2	8.4	7.4	6.5	6.6	8.3	10.5	13.2	16.0		
	5% WB	MCDBR	18.9	18.3	17.6	17.4	18.0	18.9	19.5	20.8	19.0	19.5	18.6	19.4		
		MCWBR	14.1	12.5	10.7	9.1	7.8	7.2	6.9	6.9	7.2	9.8	11.9	15.5		
(40) Clear-Sky Solar Irradiance	taub		0.332	0.356	0.371	0.427	0.476	0.466	0.490	0.475	0.446	0.376	0.370	0.339		
	taud		2.514	2.460	2.411	2.257	2.157	2.237	2.197	2.239	2.281	2.462	2.425	2.501		
	Ebn at Noon		286	288	290	276	262	263	256	259	263	277	269	277		
	Edh at Noon		28	32	36	44	49	45	46	44	41	32	30	27		
(44) All-Sky Solar Radiation	RadAvg		907	1013	1389	1684	1850	1951	1956	1864	1595	1350	1012	792		
	RadStd		86	187	125	109	139	159	157	138	152	175	105	98		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (36 neighbors)		+0.61	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+198	+171

Nomenclature: See separate page